

the assumption used to build the ladder, leading the real value of the cash flows to fluctuate in unpredictable ways. Only TIPS can hedge unexpected inflation outside of what is assumed for the COLA.

The exhibit shows the cumulative cost (using wholesale prices without any potential markups in pricing for retail consumers) of building \$40,000 with 2 percent growth for up to thirty years. To get a clearer understanding of how the exhibit works, consider funding the income goal for 2027. With a 2 percent COLA, you seek to generate \$48,760 that year. You want maturing bonds with this face value. The yield on a strip maturing that year is 2.48 percent. Discounting the spending amount to the present, you must set aside \$37,256 in the bond today with a 2.48 percent annual return to have \$48,760 available from the maturing bond in 2027. The same process works for every date income is desired.

Over the full thirty years, the total ladder cost is \$1.045 million. This implies an initial spending rate of 3.83 percent for the \$40,000 initial spending goal and an implied internal rate of return of 2.8 percent on the cash flows supported through the initial ladder cost. In other words, 2.8 percent is the fixed investment return required for the initial \$1.045 million payment to support the subsequent cash flows over the next thirty years. This is the maximum spending that can be supported by the Treasury STRIPS' yield curve in the current market environment. Efforts to spend at a higher rate with risky investments like stocks will create risk that the goal cannot be met.

Exhibit 7.11

Constructing Retirement Income Bonds Ladders Using Treasury STRIPS on January 3, 2017, for an Annual Income of \$40,000 with a 2% Annual Cost-of-Living Adjustment

Maturity Date	Ask Price	Yield to Maturity (%)	Targeted Spending Amount	# of Shares Purchased	Cost of Shares	Cumulative Cost to Build Bond Ladder of This Length	Implied Portfolio Bond Allocation for Wealth of \$1 Million
2017	\$1,000.00	n/a	\$40,000	40	\$40,000	\$40,000	4.0%